INJECTIVITY OF AN EXT GROUP ALONG INJECTIVE MODULE MAPS FORCES A SPLITTING OF THE TRUNCATION TRIANGLE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author writes '(this is where we use that K is bounded above)' inside the counted proof span, reproduced verbatim. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Truncation distinguished triangles and the computation of Ext groups of a truncation. invoked at proof lines 19929-19939. Supplies the exact sequence whose last term is Hom_R(H^a(K), M).
- Vanishing of Ext of a bounded above complex out of an injective module, and the fact that every module embeds in an injective one. used at proof lines 19948-19952. Is what turns the hypothesis about injective maps into vanishing for all modules.
- The criterion that vanishing of Ext^{-a} against all modules is equivalent to projective amplitude in [a+1, b], and the unique-splitting lemma. invoked at proof lines 19953-19957. Produces the amplitude bound and then the unique direct sum decomposition.

Source reference. Original lemma 06XR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8860–8860. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06XS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8861–8861. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08J5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10838–10838. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 17414: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17414–17414. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19704: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19704–19704. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $K \in D^-(R)$. Let $a \in \mathbf{Z}$. Assume that for any injective R-module map $M \to M'$ the map $\mathrm{Ext}^{-a}_R(K, M) \to \mathrm{Ext}^{-a}_R(K, M')$ is injective. Then there is a unique direct sum decomposition $K \cong \tau_{\leq a}K \oplus \tau_{\geq a+1}K$ and $\tau_{\geq a+1}K$ has projective-amplitude in $[a+1, b]$ for some b.*

Proof. Consider the distinguished triangle

$$\tau_{\leq a}K \to K \to \tau_{\geq a+1}K \to (\tau_{\leq a}K)[1]$$

in $D(R)$, see Derived Categories, Remark 08J5. Observe that $\mathrm{Ext}^{-a}_R(\tau_{\leq a}K, M) = \mathrm{Hom}_R(H^a(K), M)$ and $\mathrm{Ext}^{-a-1}_R(\tau_{\leq a}K, M) = 0$, see Derived Categories, Lemma 06XS. Thus the long exact sequence of Ext gives an exact sequence

$$0 \to \mathrm{Ext}^{-a}_R(\tau_{\geq a+1}K, M) \to \mathrm{Ext}^{-a}_R(K, M) \to \mathrm{Hom}_R(H^a(K), M)$$

functorial in the R-module M. Now if I is an injective R-module, then $\mathrm{Ext}^{-a}_R(\tau_{\geq a+1}K, I) = 0$ for example by Derived Categories, Lemma 06XR. Since every module injects into an injective module, we conclude that $\mathrm{Ext}^{-a}_R(\tau_{\geq a+1}K, M) = 0$ for every R-module M. By Lemma 17414 we conclude that $\tau_{\geq a+1}K$ has projective-amplitude in $[a+1, b]$ for some b (this is where we use that K is bounded above). We obtain the splitting by Lemma 19704. □